

3.2.11 Robust and secure programming

Content	Additional information
Be able to write simple data validation routines.	Students should be able to use data validation techniques to write simple routines that check the validity of data being entered by a user. The following validation checks are examples of simple data validation routines: • checking if an entered string has a minimum length • checking if a string is empty • checking if data entered lies within a given range (eg between 1 and 10).
Be able to write simple authentication routines.	Students should be able to write a simple authentication routine that uses a username and password. Students will only be required to use plain text usernames and passwords (ie students will not need to encrypt the passwords).
Understand what is meant by testing in the context of algorithms and programs. Be able to correct errors within algorithms and programs.	
Understand what test data is and describe the following types of test data: • normal (typical) • boundary (extreme) • erroneous data.	Boundary data would be for example: If the allowed range is 1 to 10, then boundary data is 0, 1, 10, 11, ie either side of the allowed boundary.
Be able to select and justify the choice of suitable test data for a given problem.	
Understand that there are different types of error: • syntax error • logic error.	
Be able to identify and categorise errors within algorithms and programs.	